Intracranial Hypertension & Herniation

CLINICAL SIGNS OF HERNIATION

- **Motor decline:** Spontaneous GCS motor score decrease of ≥ 1 point.
- **Pupillary reactivity:** Decrease in pupillary reactivity (Neurological Pupil Index, NPi < 3).
- **Asymmetry:** New pupillary asymmetry or unilateral dilation (ipsilateral mydriasis).
- **Focal deficit:** New focal motor deficit or abnormal posturing (decorticate / decerebrate).
- **Cushing's Triad (Late Sign):** Systolic hypertension, bradycardia, and irregular respirations. ***Cushing triad is a LATE sign of brainstem compression.***

Syndrome	Anatomical Substrate	Clinical Exam & Diagnostic Trap
Uncal (Lateral)	Medial temporal lobe (uncus) pushed over tentorial edge	Ipsilateral sluggish/dilated pupil (CN III compressed), contralateral hemiparesis. **Kernohan's Notch** causes false-localizing ipsilateral hemiparesis.
Central (Axial)	Downward diencephalic and midbrain displacement	Progressive stupor, midpoint fixed pupils, decorticate to decerebrate posturing. Symmetrical signs often confused with metabolic encephalopathy.
Subfalcine (Cingulate)	Cingulate gyrus displaced under the falx cerebri	Often clinically silent, or presents with contralateral lower extremity weakness. **ACA compression** causes frontal/leg territory infarction.
Tonsillar (Downward)	Cerebellar tonsils forced through the foramen magnum	Cushing's triad, flaccid quadriplegia, respiratory arrest.

MANAGEMENT (TIERED PATHWAY)

Tier 0: Fundamental Optimization (For all at-risk patients)
- Elevate HOB 30°; strict neutral midline neck alignment to preserve venous outflow.
- Euvolemia (isotonic saline; avoid hypotonic D_5W), normothermia (treat fevers), normocapnia (pCO_2 35–40 mmHg).

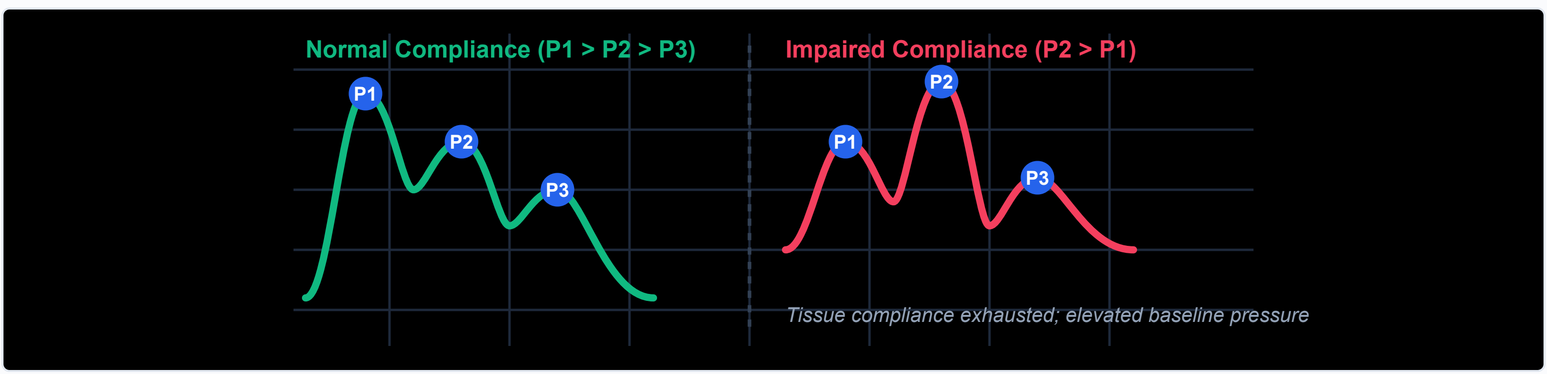
Tier 1: Initial Interventions
- Continuous CSF drainage via EVD (typically set at 10–15 cmH_2O).
- Analgesia/sedation (propofol/fentanyl) to prevent coughing, agitation, or ventilator dyssynchrony.

Tier 2: Hyperosmolar & Ventilatory Escalation
- **Mannitol 20%:** 1 g/kg IV bolus over 20-30 min. Must use in-line filter. **Hold if Serum Osmolarity > 320 mOsm/kg or Gap ≥ 20 mOsm/kg.**
- **Hypertonic Saline (HTS):** 3% (150-250 mL bolus) or 23.4% (30 mL rescue bolus; central line access only). **Hold if Serum Na > 155 mEq/L or Cl > 115 mEq/L.**
- **Controlled Hyperventilation:** Target pCO_2 30-35 mmHg. Temporary bridge only; avoid prolonged use or $pCO_2 < 30$ (ischemia risk).

Tier 3: Refractory & Salvage Measures
- High-dose barbiturate therapy (pentobarbital) titrated to burst suppression on EEG.
- **Decompressive Surgery:**
  - **Malignant MCA (DHC):** Age ≤ 60 years, GCS decline ≥ 1 / pupil changes, infarct ≥ 50% MCA, within 48h (DECIMAL/DESTINY).
  - **Cerebellar Stroke:** Suboccipital craniectomy for brainstem compression, 4th ventricle effacement, or hydrocephalus.

TIER 0: BASE hob 30°, midline, isotonic	**TIER 1: INITIAL** evd drainage, sedation	**TIER 2: MEDICAL** osmotherapy, bridge vent	**TIER 3: SALVAGE** barb coma, craniectomy

ICP WAVEFORM ANALYSIS



- **P1 (Percussion wave):** Arterial pulsation.
- **P2 (Tidal wave):** State of intracranial compliance (elastic reserve).
- **P3 (Dicrotic wave):** Venous pulsations.
- **Compliance States:**
  - **Normal Compliance:** P1 > P2 > P3 (elastic brain tissue easily cushions pulsations).
  - **Impaired Compliance / High ICP:** P2 > P1 (brain tissue reserve exhausted; high risk of herniation).

CPP = MAP - ICP
In patients with intracranial hypertension or mass effect, cerebral perfusion is highly pressure dependent, and cautious BP lowering is recommended.